

Part C: Data Request

To develop a model that can identify patients at increased risk of 30 day hospital readmission, STADIOcare should provide historical patient, hospital encounter, clinical, and discharge data. The requested data should cover a sufficiently long period to capture different patient groups and admission patterns. Where possible, data should be provided at patient and admission level using a unique, anonymised patient identifier.

1. Patient and Demographic Data

Column name	Data type	Example	Additional Information
Patient ID	String	P122445	Unique anonymised identifier;
Age	Integer	71	Age of the patient at the time of admission
Gender	Categorical	Female	
Chronic conditions	Categorical	Diabetes; Hypertension	Existing chronic conditions available before discharge

2.2. Hospital Admission Data

Column name	Data type	example	Additional Information
Admission ID	String	A122	Unique identifier for each hospital encounter.
Admission date	Date/time	2026-03-14 09:25	Date and time of admission.
Discharge date	Date/Time	2025-03-18 14:10	Date and time of discharge.
Admission type	Categorical	Emergency	Indicate whether the admission was,emergency, elective,Birth,surgery,or another applicable category.
Procedure codes	Categorical	PROC1023	Procedures performed during the encounter
Discharge disposition	Categorical	Home	Patient destination or status at discharge.
Previous readmissions	Integer	1	Number of previous readmissions, with the relevant time period specified.
Previous outpatient visits	integer	3	Number of outpatient or clinic visits within a defined historical period.

3. Clinical and Treatment Data

Column name	Type	Example	Additional Information
Primary diagnosis	Categorical	Type 2 Diabetes	Main diagnosis for the admission
Number of diagnoses	Integer	5	Total recorded diagnoses for the encounter.
Number of medications	Integer	7	Number of medications associated with the admission

4. Readmission Outcome Data

Column name	Data type	Example	Additional Information
Readmission within 30 days	Binary/categorical	1=yes /0=No	Target variable: 1/Yes if the patient is readmitted within 30 days after discharge, otherwise 0/No.
Readmission date	Date/Time	2025-04-02 11:30	Date and time of the subsequent admission, where applicable
Readmission type	Categorical	Emergency	Identify whether the readmission was emergency, planned, or another applicable category.
Readmission diagnosis	Categorical	Heart failure	Primary diagnosis associated with the readmission

Additional Data Requirements

The dataset should include clear definitions for each variable, consistent coding across hospitals, and information about missing or unknown values. The readmission outcome should be clearly defined, and planned readmissions should be identifiable so they can be considered separately from potentially unplanned returns to hospital.